NLA Midterm (November 2025) Variant B

Problem 1. (1 + 2 pts)

(a) Find a rotation matrix A of the (x, y) plane in $\mathbb{R}^3$ with $A^5 = I$.

(b) Find the eigendecomposition of A.

Problem 2. (2 + 1 pts)

$$A = \begin{bmatrix} 0.7 & 0.3 \\ 0.2 & 0.8 \end{bmatrix}$$

(a) Find the eigendecomposition of A.

(b) Compute $\lim_{k\to\infty} A^k$.

Problem 3. (3 + 1 pts)

$$A = \begin{bmatrix} 1/18 & 2/9 & 13/9 \\ 4/9 & 1/36 & 14/9 \\ 2/9 & -1/9 & -11/9 \end{bmatrix}$$

(a) Find the QR decomposition of A (with pivoting if necessary).

(b) Find the Skeleton decomposition of A.

Problem 4. (1 + 3 pts)
Given the data points $(2, 1)$, $(0, 1)$, $(-2, 1)$:

(a) Formulate the least squares problem to find the best-fit line.

(b) Find the minimum ℓ_2-norm least squares solution.

Problem 5. (2 pts)
Let $A \in \mathbb{C}^{n\times n}$ and $M, L \subset \mathbb{C}^n$ be m-dimensional subspaces such that A is positive definite and $M = L$. Let $W, V \in \mathbb{C}^{n\times m}$ be bases of M and L. Show that $W^*AV \in \mathbb{C}^{m\times m}$ is non-singular.

Problem 6. (3 + 1 pts)

(a) Let $A \in \mathbb{C}^{n\times n}$ be positive definite, $b, x_0 \in \mathbb{C}^n$, and let $M \subset \mathbb{C}^n$ be a subspace. Show that a vector x satisfies

$$x \in x_0 + M, \qquad b - Ax \perp M$$

if and only if

$$x = \arg\min_{y\in x_0+M} \|A^{-1}b - y\|_A.$$

(b) Show that

$$A^{-1}b - x = (I - P_A)(A^{-1}b - x_0),$$

where P_A is the orthogonal projector onto the subspace M w.r.t the A-inner product.

Comment: Recall that A-inner product is defined as $\langle x, y\rangle_A = x^*Ay$.